HACK MATRIX HACKATHON

TITLE PAGE

- **PROJECT TITLE : Hidden Gems Local Guide**

- **TEAM NAME : CODEX GLADIATORS**

- **TEAM MEMBERS : Saumya Agarwal - 25BCE10298 (Team Lead)**
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PROBLEM STATEMENT

- Describe the problem clearly: Popular travel apps primarily focus on major tourist attractions, overlooking unique, lesser-known spots that locals cherish.

- Who is affected by this problem: Travelers seeking authentic experiences, and local businesses that miss out on potential visitors.

- Why this problem is important: It leads to overcrowded mainstream tourist spots and deprives travelers of culturally rich, authentic local experiences.

EXISTING SOLUTIONS AND GAPS

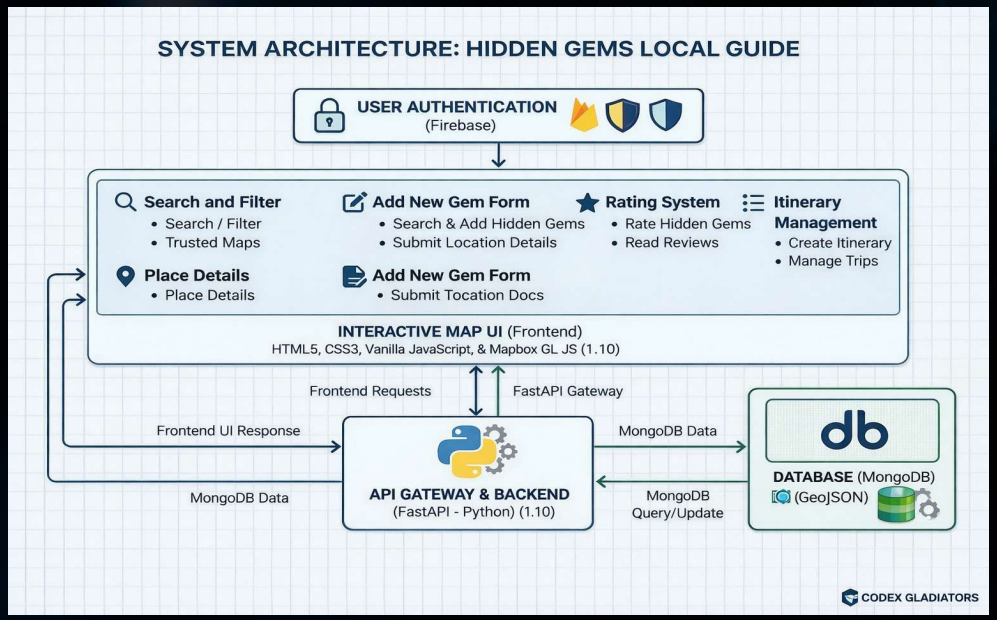
- List of existing Solutions: Google Maps, TripAdvisor, Yelp.
- Mention their Limitations: They are saturated with popular tourist traps, lack curated 'local-only' perspectives.
- Explain the gap your project will address: We will provide a community-driven platform exclusively dedicated to uncovering and verifying local-approved hidden gems.

PROPOSED SOLUTION

- Explain your solution clearly: A crowdsourced platform where verified locals can submit hidden gems, and visitors can explore them on an interactive map.
- Explain how it solves the problem: It directly connects travelers with local knowledge, steering them away from tourist traps to authentic experiences.
- List key features: Interactive map, custom pins, place detail pages, community submission form, user ratings.

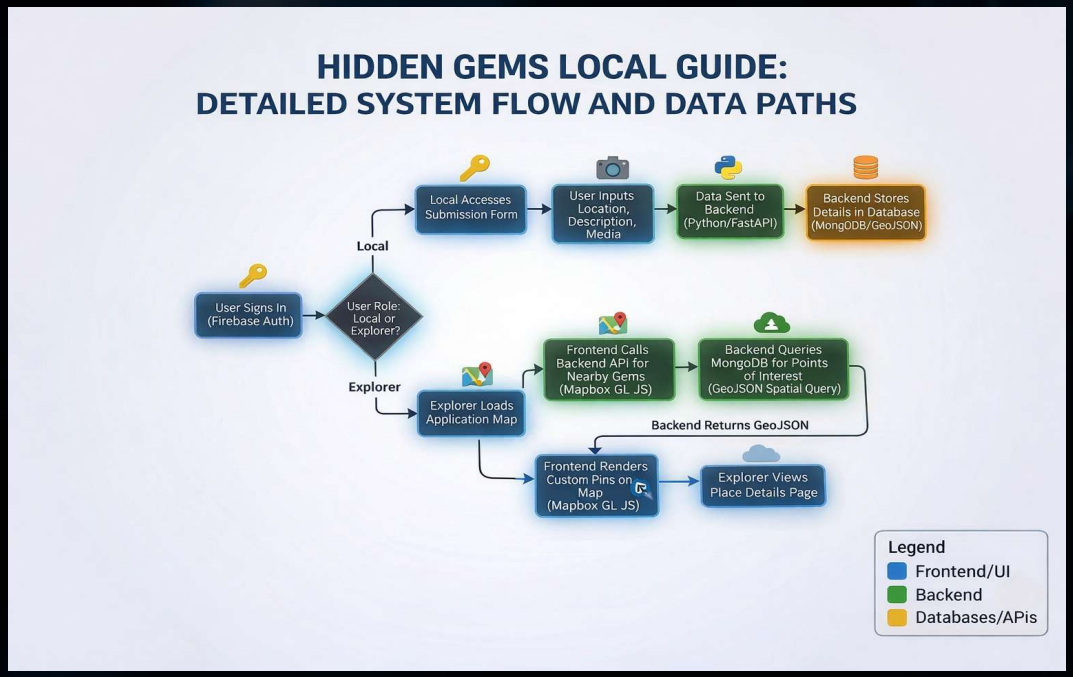
SYSTEM ARCHITECTURE

- Components: Frontend (HTML5, CSS3, and Vanilla JavaScript), Backend (Python with FastAPI), Database (MongoDB), APIs (Mapbox GL JS)
- Explain data flow briefly: User requests map -> API Gateway -> Backend retrieves locations from DB -> Spatial coordinates rendered on Map UI.

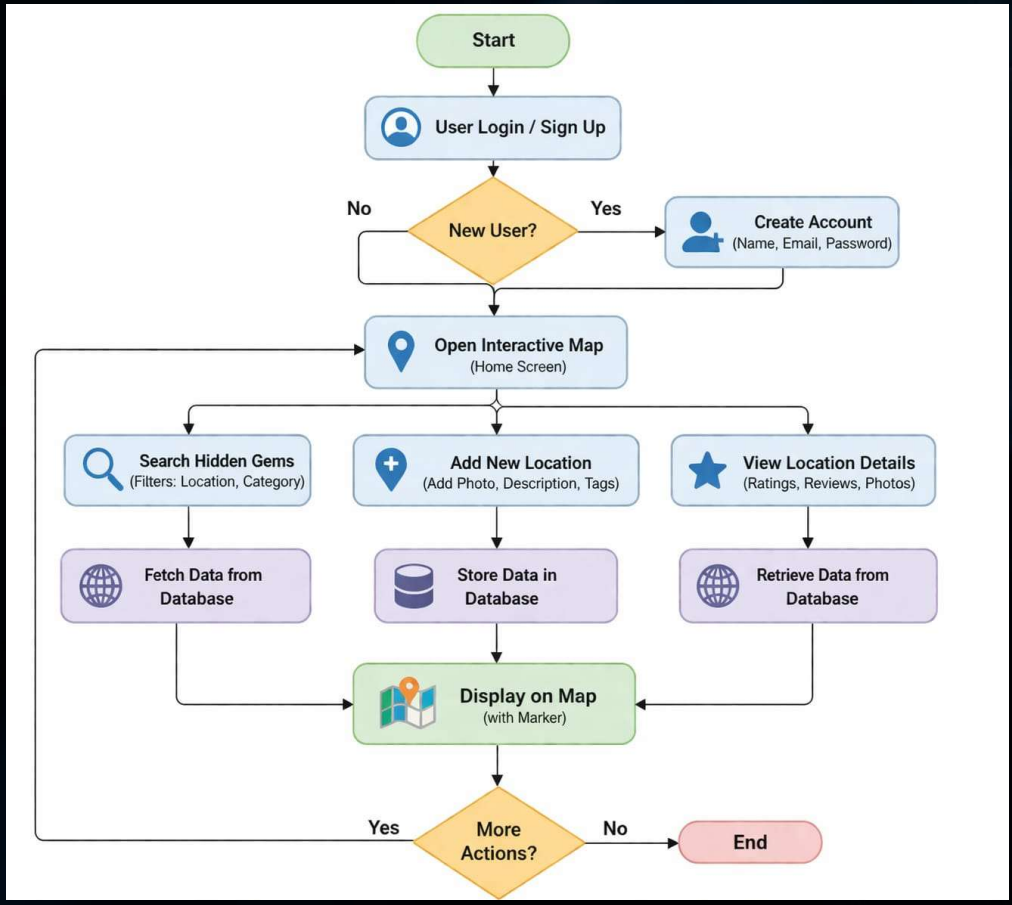


WORKING PROCESS

- Show step-by-step working: 1. User signs in
  2. Local adds location with media
  3. Explorer browses interactive map and views details



FLOWCHART


Start
User Login / Sign Up
No
New User?
Yes
Create Account
(Name, Email, Password)
Open Interactive Map
(Home Screen)
Search Hidden Gems
(Filters: Location, Category)
Add New Location
(Add Photo, Description, Tags)
View Location Details
(Ratings, Reviews, Photos)
Fetch Data from Database
Store Data in Database
Retrieve Data from Database
Display on Map
(with Marker)
Yes
More Actions?
No
End

TECH STACK

- Frontend technologies : HTML5, CSS3, and Vanilla JavaScript
- Backend technologies : Python with FastAPI
- Database: MongoDB (GeoJSON support)
- Tools and APIs: Mapbox GL JS(for rendering interactive map and custom pins), Firebase (authentication and sync).

IMPLEMENTATION PLAN

- Step-by-step development plan: 1. UI/UX & DB Schema
  2. Backend API & Map integration
  3. Frontend dev
  4. Testing & deploy

- Modules to be built: User Auth, Interactive Map, Submission Form, Reviews & Rating

EXPECTED OUTCOME & IMPACT

- Expected results of the project: A functional prototype featuring an interactive map with custom pins, detailed place pages, and submission form.
- Benefits to users: Travellers get authentic experiences; locals showcase their community.
- Real-world impact: Promotes sustainable tourism by distributing foot traffic away from congested hotspots to hidden gems.
- AI Fake Detection System: The Platform will use AI to detect fake, misleading or overhyped locations using AI.

FUTURE SCOPE AND CONCLUSION

- Future improvements: AI-powered personalized itineraries, gamification for contributors (badges/rewards).
- Final summary: Hidden Gems redefines travel by empowering locals to share their world, offering tourists authentic journeys.
- Scalability of the project: Cloud-native architecture handles multiple cities and countries as user base expands.